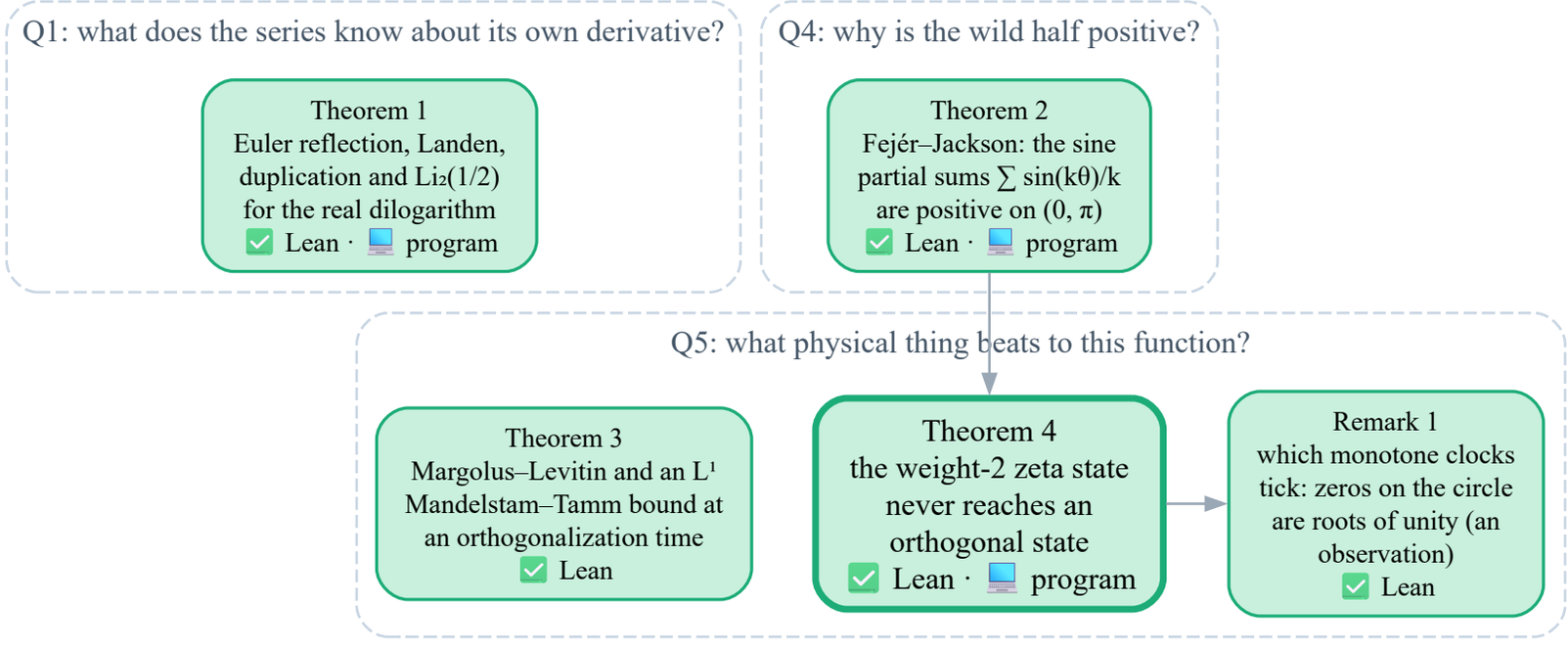
# Map of *The Clock That Never Ticks: a machine-checked path from the dilogarithm to quantum speed limits in Lean 4*

Special functions · the dilogarithm and the Clausen function

**5** results | ✅ **5** entirely in Lean | ◐ **0** partly in Lean | 💻 **3** checked by a program | written proof only | (a result can carry several tags)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



## What the paper contributes

A sorry-free Lean 4 path from the dilogarithm, which Mathlib cites but does not define, to a quantum state that never orthogonalizes.

- **The real dilogarithm** with Euler's reflection, Landen's transformation, the duplication formula and $\mathrm{Li}_2(1/2)$ (Theorem 1); the golden-ratio values and the Lee–Yang $c_{\mathrm{eff}} = 2/5$ from a 3×3 linear system, without the five-term relation.
- **The circle**: the Clausen function and Catalan's constant $G = \mathrm{Cl}_2(\pi/2)$; the Fejér–Jackson inequality (Theorem 2) and, by Abel summation without asymptotics, $\mathrm{Cl}_2(\theta) \geq \sin(\theta)/2$ on $(0, \pi)$.
- **Speed limits**: Margolus–Levitin and an $L^1$ form of Mandelstam–Tamm for a scalar spectral measure (Theorem 3).
- **The assembly** (Theorem 4): populations $6/(\pi^2 n^2)$ on levels $E_n = n$ give infinite mean energy and variance, so both bounds are silent, and yet the state never reaches an orthogonal state.
- The mathematics is classical; the paper claims the formalization and the assembly, and records the Eneström–Kakeya reading as an observation (Remark 1).

🚀 **Start here:** **Theorem 1**: Euler reflection, Landen, duplication and $\mathrm{Li}_2(1/2)$ for the real dilogarithm.

**Main results:**

- **Theorem 4**: the weight-2 zeta state never reaches an orthogonal state

## 🧭 Fields: where the paper lives and what it connects

**Tools taken from other branches**

- **Power series and real analysis** → term-by-term differentiation, the Weierstrass M-test, mean-value constancy, the boundary limit $\ln x \cdot \ln(1-x) \to 0$ — Theorem 1
- **Fourier series and positive trigonometric sums** → Mathlib's Bernoulli–Fourier expansion for the cosine half; a division-free Dirichlet telescope and an interior-minimum induction; Abel summation without asymptotics — Theorem 2, Theorem 4
- **Quantum mechanics** → speed limits stated for a scalar autocorrelation of a discrete spectral measure, with the cosine inequality $\cos x \geq 1 - (2/\pi)(x + \sin x)$ — Theorem 3

**What the results serve**

- **Quantum speed limits** ← a state of infinite mean energy and variance that never orthogonalizes: both textbook bounds are silent on it — Theorem 4
- **Dilogarithm identities** ← the golden-ratio values and the Lee–Yang c_eff = 2/5 without the five-term relation; the five-term relation is the companion paper — Theorem 1
- **Roots of positive-coefficient series (Eneström–Kakeya)** ← an observation (formalized after publication, for series too): which monotone states orthogonalize — 1